Lab Thirteen – Statistical Inference

1 Remaining Portfolio Work!

By 5p 4/27 Email me a link to your GitHub page with all polished labs rendered, and your review material (at least one unit). In the email, you should include a reflection about the processes involved in creating this portfolio and what you learned as a result. If you're having trouble reflecting, consider Fink's Taxonomy of Significant Learning (Google Fink taxonomy aligned self-reflection questions). In the email, let me know whether you would be comfortable sharing your page – I'm hoping that together we can get a good study guide for the final.

2 Statistical Inference

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: https://www.nature.com/articles/s41586-025-08729-1.

Note that the measurement is rather complicated. First, the measurements are taken on syllables of a bird song. When discussing bird songs, the term syllable refers to a single, distinct, uninterrupted sound that is part of a possibly complex bird song.

They initially describe their measurement as follows. They compare the sound of the developing zebra finches to the median sound of adult zebra finches, that act as a "goal" sound.

> "For each syllable rendition, we calculated the proximity (Euclidean distance) to the median of the adult renditions of the syllable (age > 90 days) in feature space (Fig. 2a,b)."

They contextualize that measurement by computing a "relative quality" that takes into account their current ability (e.g., are they getting closer or further compared to their previous 14 renditions).

> "Because dopamine is hypothesized to encode the relative value of performance, we defined the 'relative quality' of a syllable as proximity to the adult version of the syllable relative to the mean of the previous 14 renditions (Methods)."

They define the "closer" measurements as the 10% of syllable renditions with the highest relative quality and the "further" as the lowest 10%.

> "To test for online error responses, for each day we compared the dopamine activity associated with the
> 10

Note that the data are measures of dopamine change, they use fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers the percent change in flourescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

Finally, the researchers note that they adjust the observations to correct for the increased dopamine for singing at all (whether it is close or far).

> "As previously reported, the baseline level of dopamine in Area X is known to increase during singing; to isolate the effect of syllable quality from these singing-related dopamine increases, we subtracted the mean dopamine response across all renditions of a syllable on each day from the dopamine response for each rendition."

So, the random variables we are working with are:

$$X_{\text{close}} = \text{average \% change in fluorescence for the top 10\% syllable renditions}$$
$$\text{relative to distance from adult sound (corrected for singing)}$$
$$X_{\text{far}} = \text{average \% change in fluorescence for the bottom 10\% syllable renditions}$$
$$\text{relative to distance from adult sound (corrected for singing)}$$

That is, the unit of measure is the syllable as measured on six birds, and the measurement is somewhat complicated.

2.1 Part a

Using the `pwr` package for `R` (Champely 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

Solution

```
(power = pwr.t.test(
  n = NULL, d = 0.65, sig.level = 0.05, power = 0.80,
  type = "one.sample", alternative = "two.sided"))
```

```

     One-sample t test power calculation

              n = 20.58039
              d = 0.65
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

```
# graph if i have time
```

The researchers should have at least 21 observations.

2.2 Part b

Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the `closer_vals` and `further_vals`. Ensure to `mutate()` the data to get a difference (e.g., `closer_vals - further_vals`).

Solution

To get the data for Figure 2(g), I downloaded the source data into an unweildly and very large Excel file. Only a small subset of these data were needed. I copied the `closer.vals` data to the `further.vals` sheet, and then exported that sheet to this lab in my Mac files. Now, I have the data locally and it will be easy to keep track of in the future.

```
dat.kasdin = read.csv("CompStatLab13Data.csv")
head(dat.kasdin)
```

```
  Farther.vals Closer.vals
1  -0.22356329  0.35598310
2  -0.10703255  0.05424353
3  -0.18709396  0.14366417
4  -0.11047283  0.12247571
5  -0.17000567  0.10870189
6  -0.08973481  0.31833828
```

2.3 Part c

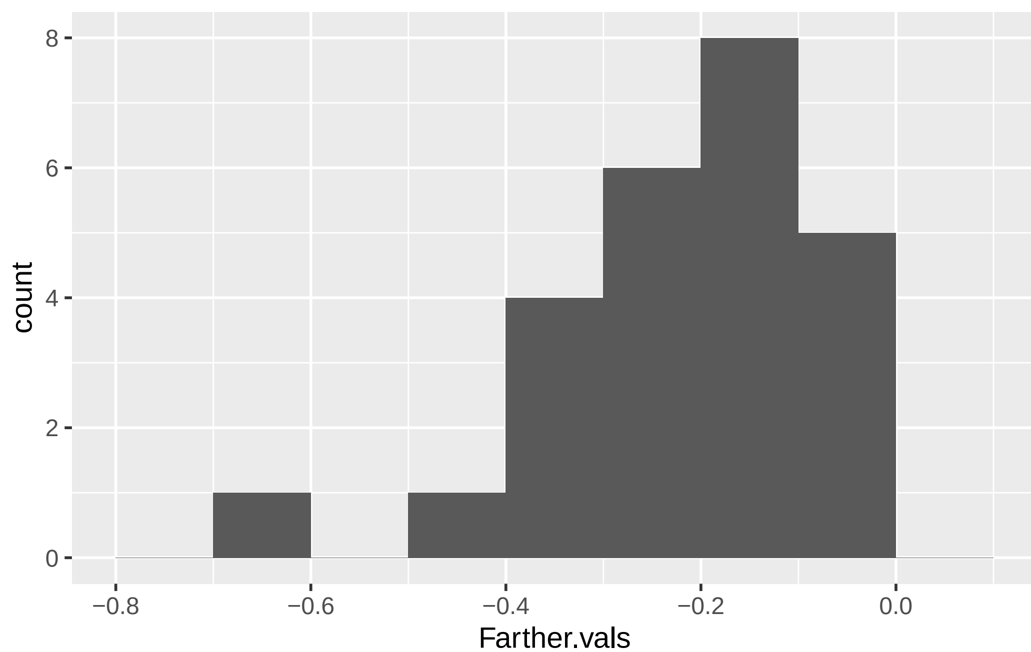
Summarize the data.

- Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?
- Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?
- Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?
- **Optional Challenge** Can you reproduce Figure 2(g)? Note that the you can use `geom_errorbar()` to plot the range created by adding the mean $\pm$ one standard deviation.

Solution

```
plot = dat.kasdin |>
ggplot(aes(x = Farther.vals)) +
  geom_histogram(breaks = seq(-0.8, 0.1, by = 0.1))
plot
```

Figure 1: Histogram of Farther Values




```
summary = dat.kasdin |>
  summarise(
    Mean = mean(Farther.vals),
    Median = median(Farther.vals),
    'Standard Deviation' = sd(Farther.vals),
    Variance = var(Farther.vals)
  )|>
knitr::kable()
summary
```

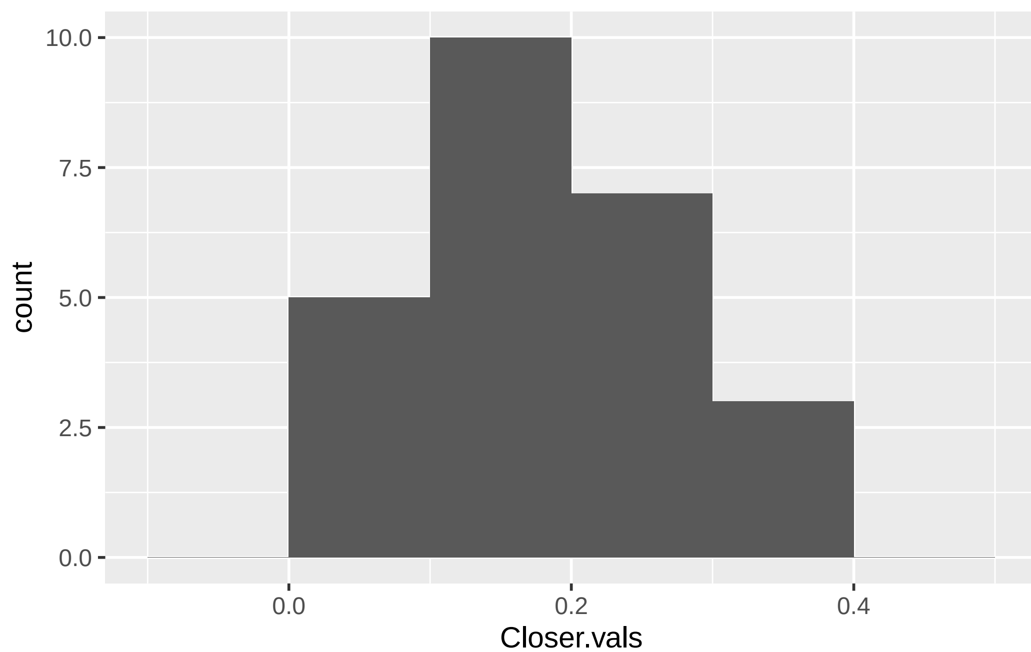
Table 1: Summary of the Farther Values

Mean	Median	Standard Deviation	Variance
-0.2128063	-0.1880146	0.1342712	0.0180287

We have preliminary evidence that supports the conjecture that dopamine in the brains of young zebra finches decreases when they sing further away from their adult song- our mean and median are below zero, and our standard deviation is less than the distance from zero of both the mean and median. The variance is very small. The histogram shows an approximately left-skewed distribution, with some values near zero.

```
plot = dat.kasdin |>
ggplot(aes(x = Closer.vals)) +
  geom_histogram(breaks = seq(-0.1, 0.5, by = 0.1))
plot
```

Figure 2: Histogram of Closer Values




```
summary = dat.kasdin |>
  summarise(
    Mean = mean(Closer.vals),
    Median = median(Closer.vals),
    'Standard Deviation' = sd(Closer.vals),
    Variance = var(Closer.vals)
  )|>
knitr::kable()
summary
```

Table 2: Summary of the Closer Values

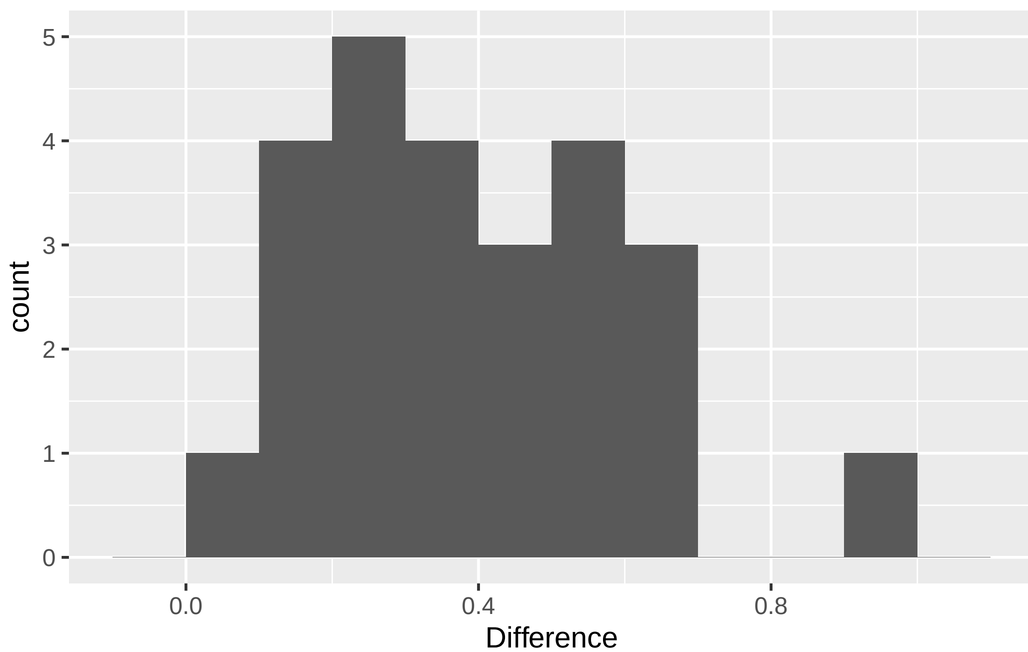
Mean	Median	Standard Deviation	Variance
0.1786241	0.1436642	0.0975315	0.0095124

We have preliminary evidence that supports the conjecture that dopamine in the brains of young zebra finches increases when they sing closer to their adult song- our mean and median are above zero, and our standard deviation is less than the distance from zero of both the mean and median. The variance is very small. The histogram show some values near zero, with an approximately symmetric distrivution.

```
dat.kasdin =  dat.kasdin |>
  mutate(Difference = Closer.vals-Farther.vals)
```

```
plot = dat.kasdin |>
ggplot(aes(x = Difference)) +
  geom_histogram(breaks = seq(-0.1, 1.1, by = 0.1))
plot
```

Figure 3: Histogram of Differences




```
summary = dat.kasdin |>
  summarise(
    Mean = mean(Difference),
    Median = median(Difference),
    'Standard Deviation' = sd(Difference),
    Variance = var(Difference)
  )|>
knitr::kable()
summary
```

Table 3: Summary of the Paired Differences

Mean	Median	Standard Deviation	Variance
0.3914303	0.3533044	0.2140658	0.0458241

We have preliminary evidence that supports the conjecture that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song- our mean and median are above zero, and our standard deviation is less than the distance from zero of both the mean and median. The variance is very small. The histogram shows an approximately uniform distribution, with one slight outlier.

2.4 Part d

Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60). Ensure to verify any conditions for use.
Note: Your numbers will vary slightly because they reported two-sided test p-values instead of the specified one-sided tests.

- "The close responses differed significantly from 0 ($p = 1.63 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The far responses differed significantly from 0 ($p = 5.17 \times 10^{-8}$).''
  Note: Use the appropriate one-sided test.
- "The difference between populations was significant ($p = 1.04 \times 10^{-8}$).''
  Note: Use the two-sided test.

Solution

For the closer values: we have a sample size of 25 and our data are approximately left-skewed. We really don't fit the conditions for use, but we are close and it will be good practice, so we will proceed!

```
library(effectsize)

# t-test:

(ttest = t.test(x = dat.kasdin$Closer.vals, alternative = "greater", mu = 0, conf.level = 0.95))
```

```

    One Sample t-test

data:  dat.kasdin$Closer.vals
t = 9.1573, df = 24, p-value = 1.333e-09
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.1452511       Inf
sample estimates:
mean of x
0.1786241
```

```
# effect size:

hedges_g(dat.kasdin$Closer.vals, mu = 0, ci = 0.95, alternative = "greater")
```

```
Hedges' g |      95% CI
-----------------------
1.77      | [1.24, Inf]

- One-sided CIs: upper bound fixed at [Inf].
```

```
# get confidence interval:

(ttest = t.test(x = dat.kasdin$Closer.vals, alternative = "two.sided", mu = 0, conf.level = 0.95))
```

```

    One Sample t-test

data:  dat.kasdin$Closer.vals
t = 9.1573, df = 24, p-value = 2.665e-09
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.1383650 0.2188831
sample estimates:
```

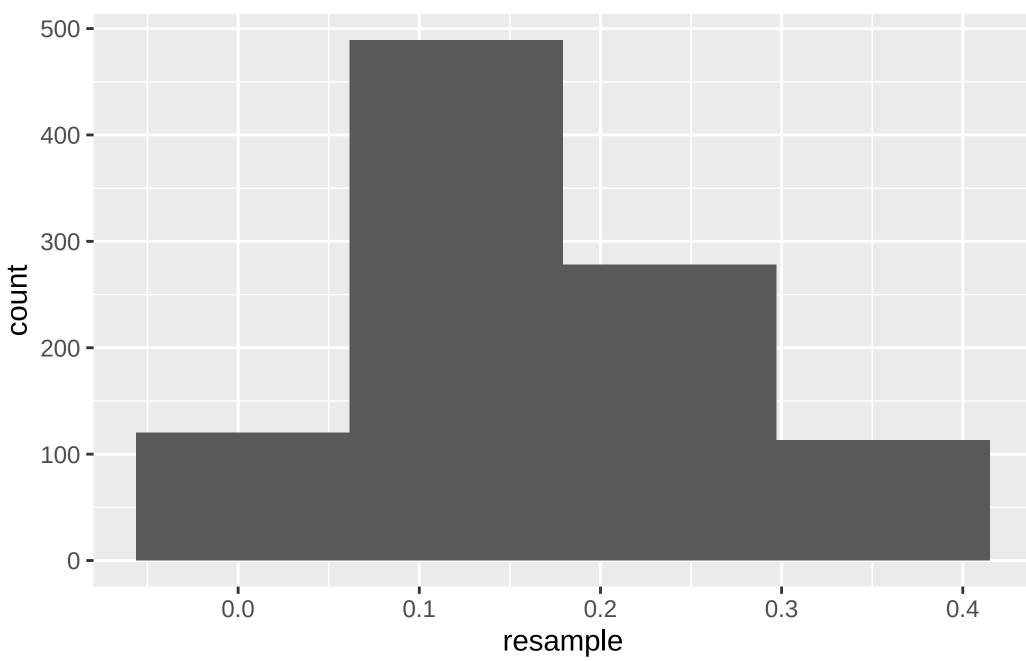
```
mean of x
0.1786241
```

We do have statistically discernible support for the hypothesis that dopamine in the brains of young zebra finches increases when they sing closer to their adult song ($t = 9.16, p < 0.05, g = 1.77, 95\% \ CI : (0.139, 0.22)$).

Now, let's check if we can still apply the central limit theorem, or if the data are too skewed. We will check using resampling!

```r
resample = sample(dat.kasdin$Closer.vals, size = 1000, replace = TRUE)

dat.plot = tibble(resample)

ggplot(dat.plot, aes(x = resample)) +
  geom_histogram(bins = 4)
```

Figure 4: histogram of the closer resampling values



Our data are still skewed. The researchers really should not have done a t-test here, as the conditions for use are not met.

For the further values: we have a sample size of 25 and our data are approximately symmetric. We have somewhat unsure footing, but we will proceed anyway!

```r
library(effectsize)

# t-test:

(ttest = t.test(x = dat.kasdin$Farther.vals, alternative = "less", mu = 0, conf.level = 0.95))
```

```
	One Sample t-test

data:  dat.kasdin$Farther.vals
t = -7.9245, df = 24, p-value = 1.866e-08
alternative hypothesis: true mean is less than 0
95 percent confidence interval:
```

```
       -Inf -0.1668619
sample estimates:
 mean of x
-0.2128063
```

```r
# effect size:

hedges_g(dat.kasdin$Farther.vals, mu = 0, ci = 0.95, alternative = "less")
```

```
Hedges' g |        95% CI
-------------------------
-1.53     | [-Inf, -1.04]

- One-sided CIs: lower bound fixed at [-Inf].
```

```r
# get confidence interval:

(ttest = t.test(x = dat.kasdin$Farther.vals, alternative = "two.sided", mu = 0, conf.level = 0.95))
```

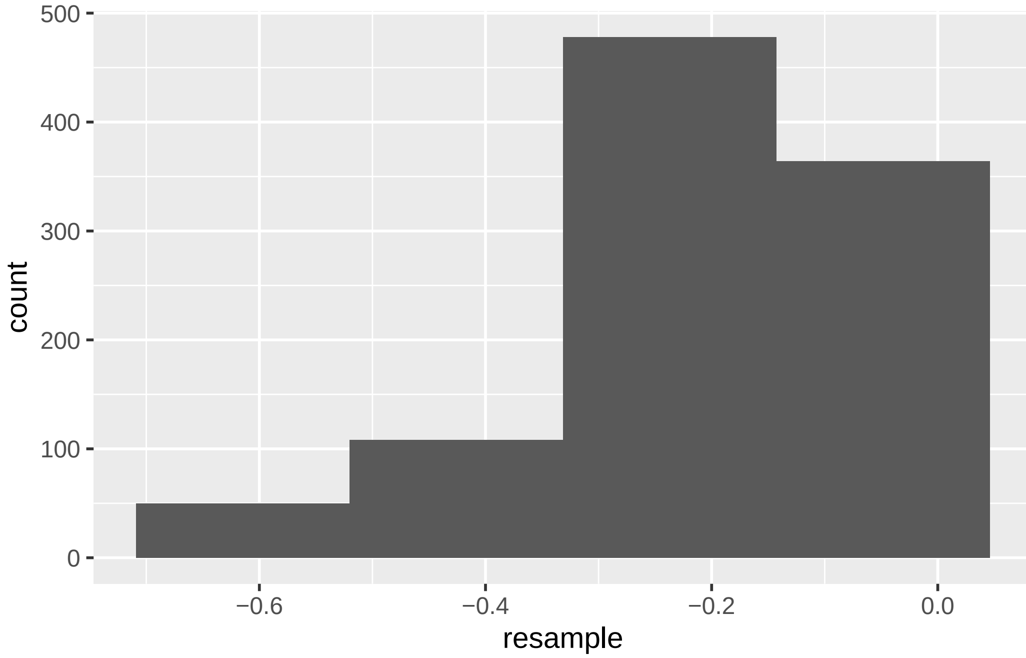
```

    One Sample t-test

data:  dat.kasdin$Farther.vals
t = -7.9245, df = 24, p-value = 3.731e-08
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.2682307 -0.1573819
sample estimates:
 mean of x
-0.2128063
```

We do have statistically discernible support for the hypothesis that dopamine in the brains of young zebra finches decreases when they sing further away from their adult song ($t = -7.92, p < 0.05, g = -1.53, 95\%\ CI : (-0.27, -0.16)$).

Now, let's check if we can still apply the central limit theorem, or if the data are too skewed. We will check using resampling!

```r
resample = sample(dat.kasdin$Farther.vals, size = 1000, replace = TRUE)

dat.plot = tibble(resample)

ggplot(dat.plot, aes(x = resample)) +
  geom_histogram(bins = 4)
```

Figure 5: histogram of the farther resampling values



Our data are still *very* skewed. The researchers really should not have done a t-test here, as the conditions for use are not met.

For the differences: we have a sample size of 25 and our data are approximately symmetric. We have somewhat unsure footing, but we will proceed anyway!

```
library(effectsize)

# t-test:

(ttest = t.test(x = dat.kasdin$Difference, alternative = "two.sided", mu = 0, conf.level = 0.95))
```

```
	One Sample t-test

data:  dat.kasdin$Difference
t = 9.1428, df = 24, p-value = 2.746e-09
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.3030683 0.4797923
sample estimates:
mean of x
0.3914303
```

```
# effect size:

hedges_g(dat.kasdin$Difference, mu = 0, ci = 0.95, alternative = "two.sided")
```

```
Hedges' g |       95% CI
------------------------
1.77      | [1.14, 2.39]
```

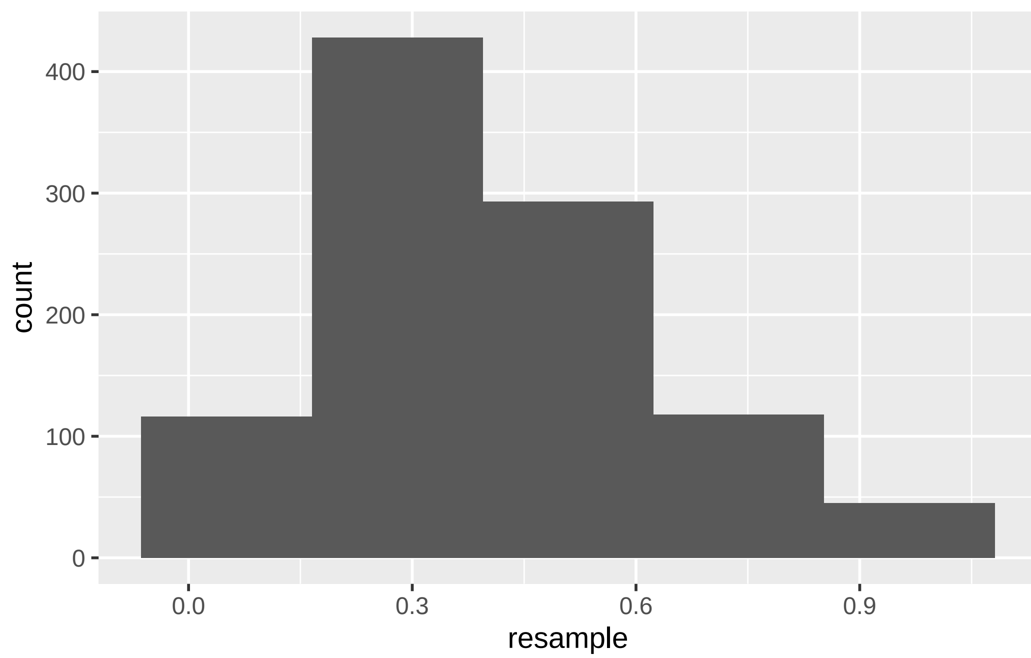
```
# Note that we already have a valid confidence interval from our original t-test.
```

We do have statistically discernible support for the hypothesis that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song ($t = 9.14, p < 0.05, g = 1.77, 95\%\ CI : (0.303, 0.480)$).

Now, let's check if we can still apply the central limit theorem, or if the data are too skewed. We will check using resampling!

```
resample = sample(dat.kasdin$Difference, size = 1000, replace = TRUE)

dat.plot = tibble(resample)

ggplot(dat.plot, aes(x = resample)) +
  geom_histogram(bins = 5)
```

Figure 6: histogram of the resampling values for the differences



Our data actually look skewed here as well. So even in this scenario, the researcher should not have done a t-test!

2.5 Part e

For each hypothesis test in the previous question, plot the null T distribution and the observed t statistic. Ensure to use `geom_ribbon()` to highlight the rejection region and the p-value. Additionally, add the resampling distribution for T to the chart to demonstrate any differences between the null distribution and the distribution suggested by the data.

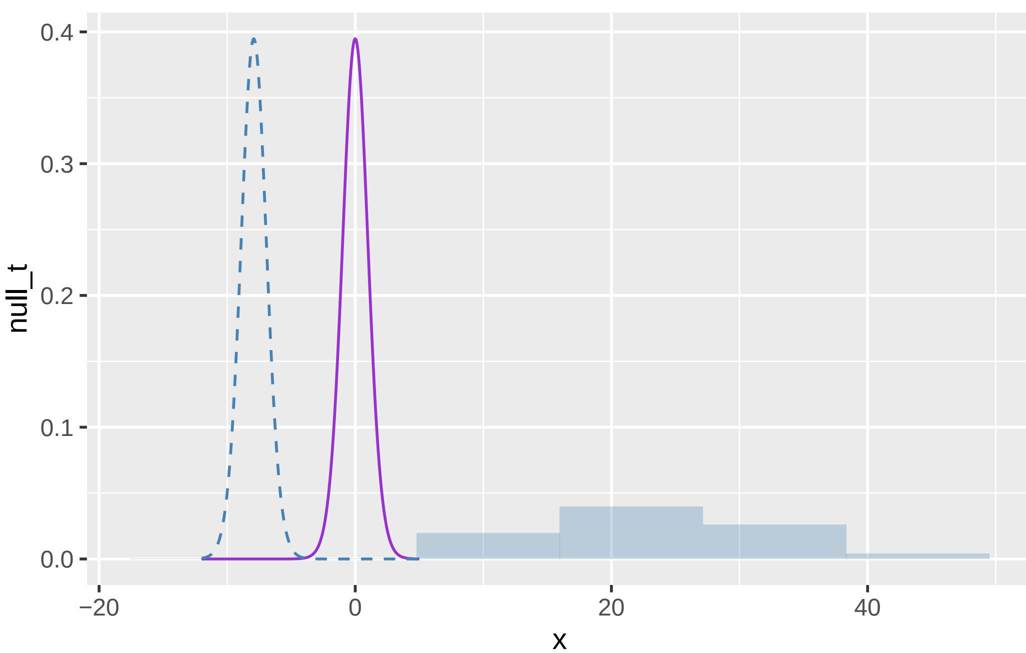
Solution

For the farther values:

```
# Let's center our t-distribution around our observed t-statistic:

t_obs <- t.test(dat.kasdin$Farther.vals, alternative = "less", mu = 0)$statistic

x <- seq(-12, 5, by = 0.05)
null_t <- dt(x, df = 24)
shifted_t <- dt(x - t_obs, df = 24)  # centered at observed t

plot.data <- data.frame(x, null_t, shifted_t)
```

```r

resample_t <- (resample - mean(dat.kasdin$Farther.vals)) /
              (sd(dat.kasdin$Farther.vals) / sqrt(nrow(dat.kasdin)))

ggplot(plot.data, aes(x = x)) +
  geom_line(aes(y = null_t), color = "darkorchid") +
  geom_line(aes(y = shifted_t), color = "steelblue", linetype = "dashed") +
  geom_histogram(data = data.frame(resample_t),
                 aes(x = resample_t, y = after_stat(density)),
                 inherit.aes = FALSE, bins = 6,
                 fill = "steelblue", alpha = 0.3)
```

Figure 7: t-distributions for the farther values and the resampling distribution



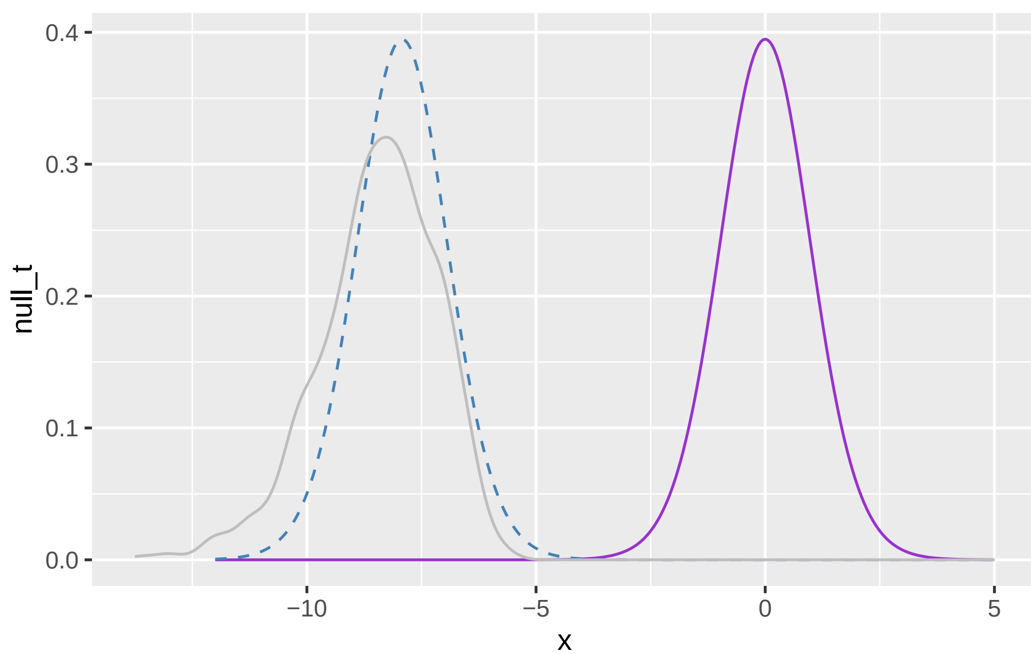
Here, we see the null t-distribution, the t-distribution shifted to be centered at our observed t-statistic, and the histogram of our resampling distribution standardized to be on the t-scale. I realize that I may have interpreted the earlier instructions for resampling wrong, and maybe it was suppoed to be bootstrapping, but I had guidance from Professor Cipolli during lab that suggested that this was correct. This histogram-scaling approach might be the correct answer, but it also isn't something I've seen before. I am including it here because I am not sure exactly why it would be incorrect (if it is incorrect), and I look forward to getting feedback!

Also, the graph with bootstrapping:

```r
# Let's center our t-distribution around our observed t-statistic:

t_obs <- t.test(dat.kasdin$Farther.vals, alternative = "less", mu = 0)$statistic

x <- seq(-12, 5, by = 0.05)
null_t <- dt(x, df = 24)
shifted_t <- dt(x - t_obs, df = 24)  # centered at observed t

plot.data <- data.frame(x, null_t, shifted_t)

boot_t_far <- replicate(1000, {
  s <- sample(dat.kasdin$Farther.vals, size = nrow(dat.kasdin), replace = TRUE)
```

```
  (mean(s) - 0) / (sd(s) / sqrt(nrow(dat.kasdin)))
})

ggplot(plot.data, aes(x = x)) +
  geom_line(aes(y = null_t), color = "darkorchid") +
  geom_line(aes(y = shifted_t), color = "steelblue", linetype = "dashed") +
  geom_density(data = data.frame(boot_t_far),
               aes(x = boot_t_far),
               inherit.aes = FALSE,
               color = "gray", alpha = 0.3)
```

Figure 8: t-distributions for the farther values



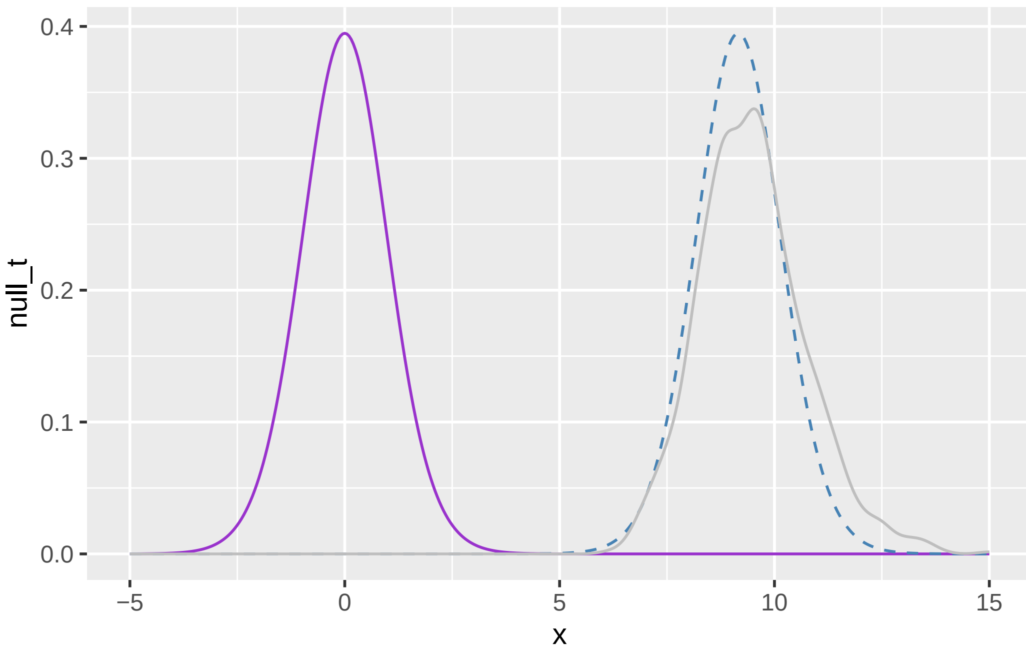
With both visualizations, the t-distribution actually appears to be an okay fit for the data. The data are a little more left-skewed, but it's not horrible.

For the closer values:

```
# Let's center our t-distribution around our observed t-statistic:

t_obs <- t.test(dat.kasdin$Closer.vals, alternative = "greater", mu = 0)$statistic

x <- seq(-5, 15, by = 0.05)
null_t <- dt(x, df = 24)
shifted_t <- dt(x - t_obs, df = 24)

plot.data <- data.frame(x, null_t, shifted_t)

boot_t_close <- replicate(1000, {
  s <- sample(dat.kasdin$Closer.vals, size = nrow(dat.kasdin), replace = TRUE)
  (mean(s) - 0) / (sd(s) / sqrt(nrow(dat.kasdin)))
})

ggplot(plot.data, aes(x = x)) +
  geom_line(aes(y = null_t), color = "darkorchid") +
```

```r
  geom_line(aes(y = shifted_t), color = "steelblue", linetype = "dashed") +
  geom_density(data = data.frame(boot_t_close),
               aes(x = boot_t_close),
               inherit.aes = FALSE,
               color = "gray", alpha = 0.3)
```

Figure 9: t-distributions for the closer values

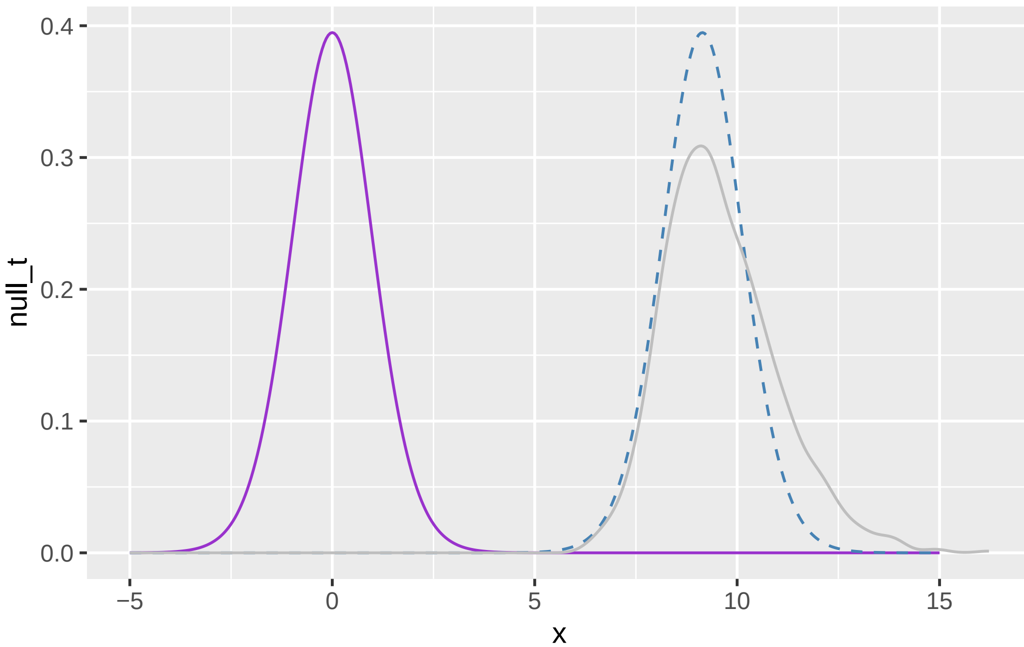


Again, for the closer values, the t-distribution actually appears to be an okay fit for the data. The data are a little more right-skewed, but it's not horrible.

For the differences:

```r
# Let's center our t-distribution around our observed t-statistic:

t_obs <- t.test(dat.kasdin$Difference, alternative = "two.sided", mu = 0)$statistic

x <- seq(-5, 15, by = 0.05)
null_t <- dt(x, df = 24)
shifted_t <- dt(x - t_obs, df = 24)

plot.data <- data.frame(x, null_t, shifted_t)

boot_t_diff <- replicate(1000, {
  s <- sample(dat.kasdin$Difference, size = nrow(dat.kasdin), replace = TRUE)
  (mean(s) - 0) / (sd(s) / sqrt(nrow(dat.kasdin)))
})

ggplot(plot.data, aes(x = x)) +
  geom_line(aes(y = null_t), color = "darkorchid") +
  geom_line(aes(y = shifted_t), color = "steelblue", linetype = "dashed") +
  geom_density(data = data.frame(boot_t_diff),
               aes(x = boot_t_diff),
               inherit.aes = FALSE,
               color = "gray", alpha = 0.3)
```

Figure 10: t-distributions for the differences



Again, for the differences, the t-distribution actually appears to be an okay fit for the data. The data are a little more right-skewed, but it's not horrible.

References

Champely, Stephane. 2020. *Pwr: Basic Functions for Power Analysis*. https://CRAN.R-project.org/package=pwr.

Kasdin, Jonathan, Alison Duffy, Nathan Nadler, Arnav Raha, Adrienne L Fairhall, Kimberly L Stachenfeld, and Vikram Gadagkar. 2025. "Natural Behaviour Is Learned Through Dopamine-Mediated Reinforcement." *Nature*, 1–8.